Non-Functional Requirements Specification: RAG-Diary

> The system's quality requirements, written as measurable quality scenarios and mapped to ISO/IEC 25010:2023 so they can be inspected and tested for real.

> Thai version: NFR-SPEC.md

1. Document control

Item	Detail
Document	Non-Functional Requirements Specification
Version	1.0
Prepared	15 October 2026
Author	Business Analyst / System Analyst (project owner)
Status	Reviewed, round 1 (25 October 2026)
Related documents	SRS, TRACEABILITY-MATRIX, TEST-PLAN, C4-ARCHITECTURE

2. Purpose and scope

This document expands NFR-01 to NFR-08 from the SRS into measurable scenarios, so the test team knows what "passing" means.

Scope: the eight quality characteristics of the system. Out of scope: functional requirements (in the SRS) and the actual test procedure (in TEST-PLAN).

3. Quality scenario format

Each requirement is written as a six-part scenario: source, stimulus, environment, artifact, response and response measure.

4. Mapping to ISO/IEC 25010:2023

NFR	Name in this system	ISO/IEC 25010:2023 characteristic
NFR-01	Performance	Performance Efficiency

NFR	Name in this system	ISO/IEC 25010:2023 characteristic
NFR-02	Security	Security
NFR-03	Privacy (PDPA)	Security, Safety
NFR-04	Usability and accessibility	Interaction Capability
NFR-05	Error handling	Reliability
NFR-06	Scalability	Flexibility
NFR-07	Maintainability	Maintainability
NFR-08	Compatibility and responsive layout	Compatibility

5. Quality scenarios

5.1 NFR-01 Performance

Part	Detail
Source	A user with an accumulated set of entries
Stimulus	The user runs a semantic search
Environment	Normal operation, with no more than 5,000 entries for that user
Artifact	The POST /search/semantic endpoint
Response	The system answers within the target time
Measure	Response time no more than 3.5 seconds at the 95th percentile

Verification: analysis and load testing. Pass criteria: p95 response time no more than 3.5 seconds.

Measured (12 Nov 2026): load test over 5,000 entries and 100 requests, p95 1.31 s (database alone 0.41 s). Pass.

5.2 NFR-02 Security

Part	Detail
Source	An attacker holding another user's token
Stimulus	An attempt to reach another user's entries
Environment	Normal operation
Artifact	Every endpoint that touches data
Response	The system refuses and returns nothing
Measure	Zero cross-user accesses, and every endpoint requires authentication

Verification: security testing. Pass criteria: no cross-user access found, and unauthenticated calls to protected endpoints return 401.

5.3 NFR-03 Privacy (PDPA)

Part	Detail
Source	The external AI provider
Stimulus	The system must send entry text for processing
Environment	The user is using a feature that needs AI
Artifact	The data flow that leaves the system boundary
Response	The system informs and asks for consent first, consent can be withdrawn in settings, and the user can export or delete their own data
Measure	Notice and consent are complete, and export and deletion work 100 percent of the time

Verification: inspection and demonstration. Pass criteria: the user consents before data leaves, and can fully manage their own data.

5.4 NFR-04 Usability and accessibility

Part	Detail
Source	A user working by keyboard or with a screen reader
Stimulus	Operating the main screens by keyboard
Environment	Current browsers
Artifact	Every main screen
Response	Everything is operable and status messages are announced
Measure	Every main path is operable by keyboard, and status messages carry an alert role

Verification: inspection and keyboard testing. Pass criteria: navigation and actions complete by keyboard, and alert regions expose a readable role.

5.5 NFR-05 Error handling

Part	Detail
Source	The external AI provider
Stimulus	An external service fails or times out
Environment	Normal operation, but the external service is unhealthy
Artifact	The screen that calls AI

Part	Detail
Response	The system shows an understandable error, never a blank or frozen screen, and the action can be retried. When image generation is blocked by quota, the page offers a mood card drawn on the device instead.
Measure	In every external failure, the screen stays usable 100 percent of the time

Verification: failure injection testing. Pass criteria: an understandable message and a retry control appear in every case.

5.6 NFR-06 Scalability

Part	Detail
Source	Growth of the entry data
Stimulus	The number of entries keeps increasing
Environment	Vector search uses an HNSW index
Artifact	The POST /search/semantic endpoint
Response	Response time stays stable
Measure	Response time does not rise sharply when the data grows tenfold

Verification: analysis and load testing. Pass criteria: the increase in response time stays within an acceptable range between a small and a large database.

Measured (12 Nov 2026): at 5,000 entries a search takes p95 0.41 s in the database and 1.31 s end to end; the extra time comes from the embedding call, not the index.

Additional measurement (12 Nov 2026): firing concurrent searches at the database layer gave p95 2.78 s with 10 concurrent users and p95 3.51 s with 30 (just past the 3.5 s target), with no failed requests. The latency is a resource limit, not a query error.

5.7 NFR-07 Maintainability

Part	Detail
Source	A developer changing the code
Stimulus	Editing or adding a function
Environment	A TypeScript codebase
Artifact	The whole codebase
Response	Type checking passes and a change happens in one place
Measure	Type checking passes with no errors, and data access comes from a single source

Verification: inspection and running the type checker. Pass criteria: the TypeScript checker reports no errors, and no data access code is duplicated.

5.8 NFR-08 Compatibility and responsive layout

Part	Detail
Source	Users on a range of devices
Stimulus	Opening the product at different screen sizes and in different browsers
Environment	Current browsers
Artifact	Every screen
Response	The layout adapts and Thai and English text render correctly
Measure	Correct rendering from phone size to large screens in current browsers

Verification: cross-browser and cross-size testing. Pass criteria: no overflow or overlap at any tested size.

6. References

Document	Relationship
SRS	The source of NFR-01 to NFR-08 and its section 8 Verification
TRACEABILITY-MATRIX	How the NFRs map to test cases TC-16 to TC-24
TEST-PLAN	The test plan that applies these scenarios
C4-ARCHITECTURE	The structure that supports performance and scalability